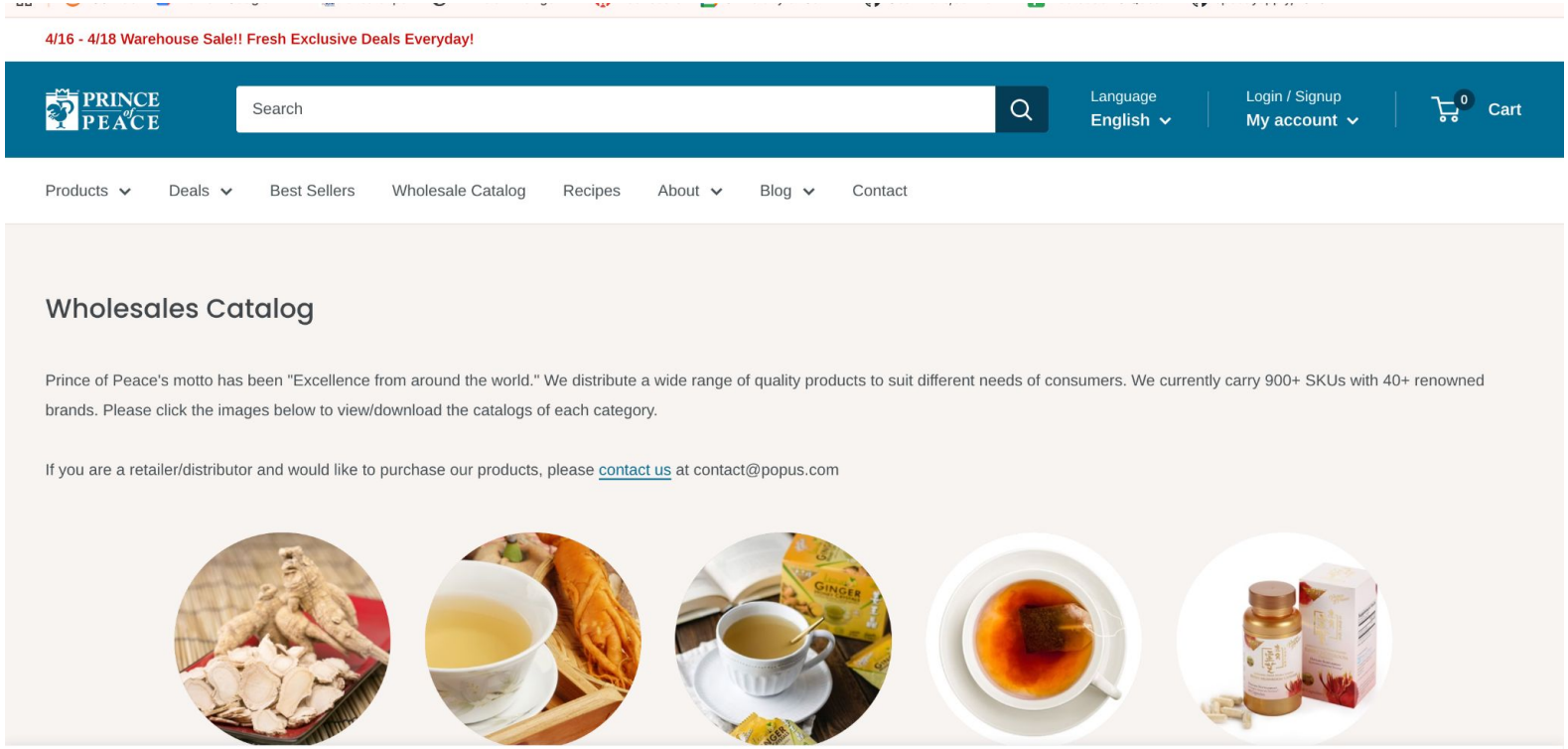
POP_off

Varshini and Nivedha

Reference Design (colors + aesthetic)



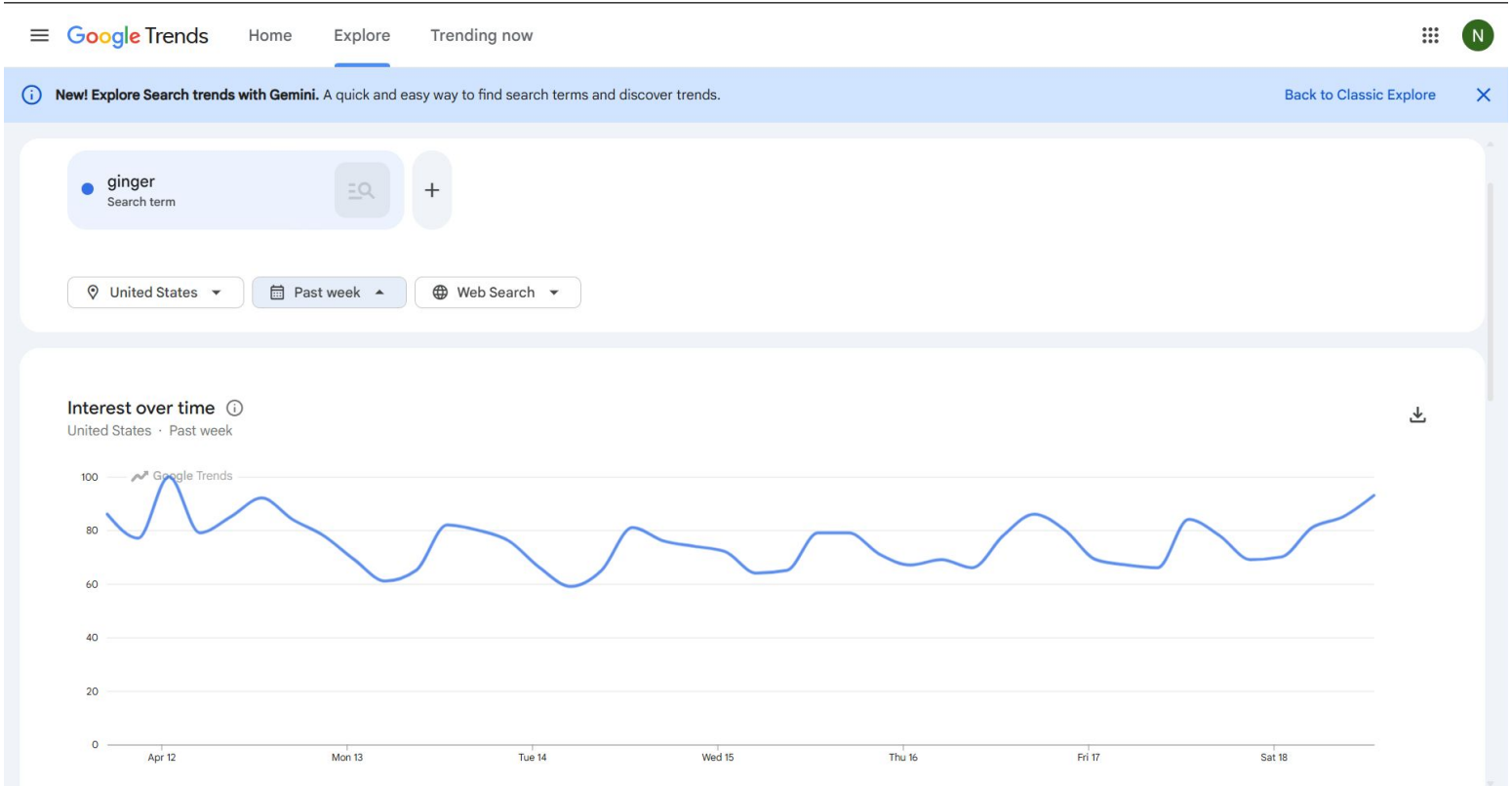
Datasets

Datasets - Real Tiktok data

74 keywords in total

Rank	Keyword	Popularity	Popularity change	CVR	CPA	Cost
1	tea	123	▲ 88.89%	3.36%	27.70 USD	1K USD
2	coffee	95	▲ 78.26%	9.82%	24.30 USD	4K USD
3	free shipping	90	▲ 91.43%	6.24%	31.90 USD	2K USD
4	50 %off	73	▲ 104.00%	2.13%	134 USD	4K USD
5	use code	69	▲ 96.15%	1.97%	75.80 USD	2K USD
6	help you	48	▲ 111.11%	2.62%	155 USD	890 USD
7	that's it	48	▲ 105.26%	2.19%	107 USD	217 USD
8	zero sugar	38	▲ 106.25%			

Datasets - Google Trends



Constraints: Hard Restrictions for filtering

datasets > restrictions > countries.txt

```
North Korea
Iran
Syria
Cuba
Belarus
Russia
Venezuela
```

datasets > restrictions > ingredients.txt

```
Dichloropropene
Azodicarbonamide
Benzophenone
Brominated vegetable oil
Calamus extract
Calamus oil
Calamus
Calcium Cyclamate
Carbon tetrachloride
Chloramphenicol
Chlorofluorocarbons
Chloroform
Cinnamyl anthranilate
Clenbuterol
Cobaltous chloride
Cobalt sulfate
Coumarin
Cyclamate
Diethyl pyrocarbonate
Diethylstilbestrol
Dihydrosafrole
Dulcin
Ethyl acrylate
Ethylene dibromide
Ethylene oxide
FD&C Orange No. 1
FD&C Orange No. 2
```

+ Shelf life > 12 months

Info: Utilized POP_ItemSpecMaster.xlsx + Synthetic Dataset

Country of Origin	Shelf Life (Months)	Maufactuer/ CoPacker	Lead Time	MOQ	Allergens
USA	60 months	ST/ QT			N/A
USA	60 months	ST/ QT			N/A
USA	60 months	ST/ QT			N/A
USA	60 months	ST			N/A
China	4 years	POP	3 months		
USA	5 years	POP			
USA	5 years	POP			
USA	5 years	POP			
USA	5 years	POP			
USA	5 years	POP			
USA	5 years	POP			
China	24 months	GL	3-4 months		
China	24 months	GL	3-4 months		
Poland	6 Months	Ferrero USA	3 weeks	2,000LB	tree nuts (hazelr
Poland	6 Months	Ferrero USA	3 weeks	2,000LB	tree nuts (hazelr
Poland	5 Months	Ferrero USA	3 weeks	2,000LB	tree nuts (hazelr
Poland	6 Months	Ferrero USA	3 weeks	2,000LB	tree nuts (hazelr
Poland	6 Months	Ferrero USA	3 weeks	2,000LB	tree nuts (hazelr
Italy	13	Loacker	3-4mths	ean container s	Contains milk, hazelnuts coco
Italy	13	Loacker	3-4mths	ean container s	Contains milk, hazelnuts,coco


```
sets > {} new_dataset.json > {} Ketchup > abc country_of_origin
{
  "Coffee Beans": {
    "keywords": [
      "coffee",
      "arabica",
      "robusta",
      "beans"
    ],
    "country_of_origin": "Ethiopia / Brazil",
    "shelf_life": "6-18 months",
    "key_ingredients": [
      "caffeine",
      "chlorogenic acids",
      "coffee oils"
    ]
  },
  "Milk": {
    "keywords": [
      "milk",
      "dairy milk",
      "whole milk"
    ],
    "country_of_origin": "Global (major production USA, India)",
    "shelf_life": "5-10 days",
    "key_ingredients": [
      "water",
      "lactose",
```


500+ entries

Signal Strength - Categorizing key opportunities and trends

Signal Strength Calculation - TikTok

- Click-through rate and conversion rate
- Cost per acquisition
- Likes & comments
- Ranking


```
[
  {
    "rank": 1,
    "keyword": "tea",
    "popularity": 123,
    "popularity_change": "88.89%",
    "ctr": "1.92%",
    "cvr": "3.36%",
    "cpa": "27.70 USD",
    "cost": "1K USD",
    "impressions": "157K",
    "6s_view_rate": "7.45%",
    "likes": "1K",
    "shares": 2,
    "comments": 2
  },
  {
```

Signal Strength Calculation - Google

- Trends over time
- Related queries


```
{
    "American Ginseng": {
        "interest_over_time": {
            "american ginseng": {
                "1744502400000": 20,
                "1745107200000": 16,
                "1745712000000": 25,
                "1746316800000": 20,
                "1746921600000": 25,
                "1747526400000": 21,
                "1748131200000": 19,
                "1748736000000": 19,
                "1749340800000": 21,
                "1749945600000": 19,
                "1750550400000": 18,
                "1751155200000": 22,
                "1751760000000": 28,
                "1752364800000": 23,
                "1752969600000": 28,
                "1753574400000": 26,
                "1754179200000": 38,
                "1754784000000": 57,
                "1755388800000": 48,
                "1755993600000": 50,
```

```
            "1775952000000": 47
        }
    },
    "related_queries": {
        "top": [
            {
                "query": "american ginseng price",
                "value": 100
            },
            {
                "query": "american ginseng benefits",
                "value": 71
            },
            {
                "query": "ginseng tea",
                "value": 45
            },
            {
                "query": "american ginseng tea",
                "value": 44
            },
            {
                "query": "american ginseng root",
                "value": 39
```

Demand vs Distribution

Distribution vs Development: POP_InventorySnapshot + PurchaseOrderHistory

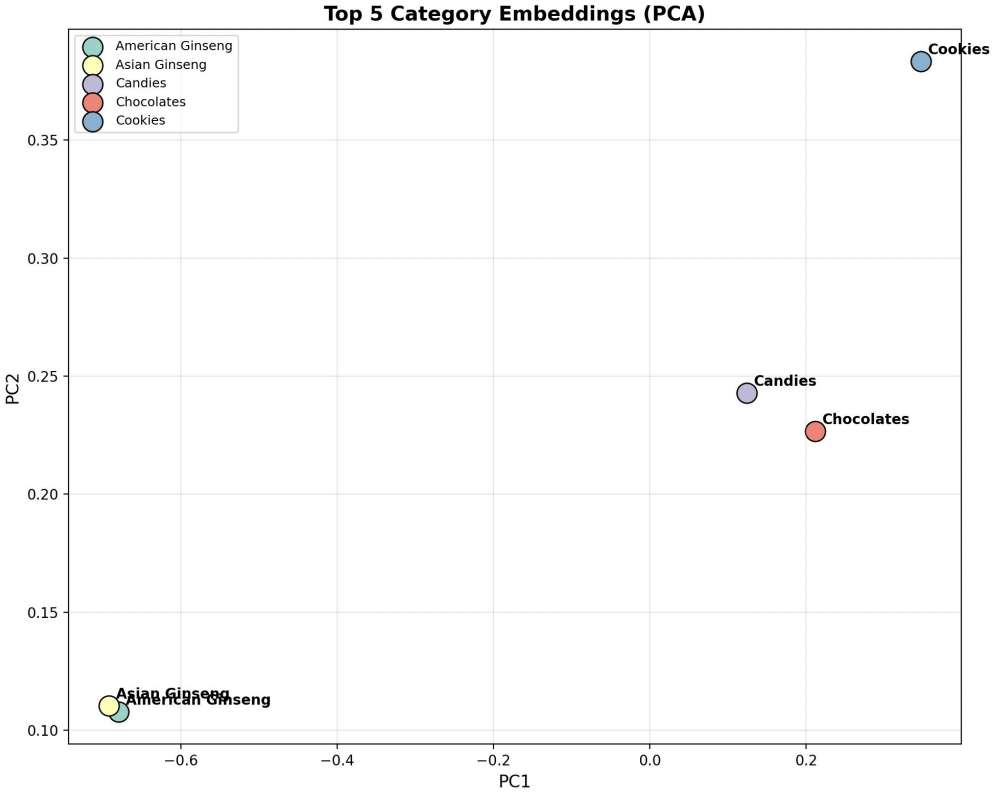
	A	B	C	D
1	Item Number	Description	Available	On Hand
2	AC-B16BK	AM GSG Root (Mixed) 1 lb Bag	4,507.	4,697.
3	AC-B3SLJ	AM GSG Slices 3 oz Bag-Jumbo	966.	1,906.
4	AC-B4BK	AM GSG Root (Mixed) 4 oz Bag	8,137.	9,550.
5	AC-B6SLJ	AM GSG Slices 6 oz Bag-Jumbo	1,216.	1,607.
6	AC-B8BK	AM GSG Root (Mixed) 8 oz Bag	6,644.	7,321.
7	AC-B9SL	POP AM GSG Root Slices 9 oz	0	0.
8	AC-B9SLE	POP AM GSG Root Slices 9 oz	0.	0.
9	A-61012	Wild AM GSG Instant Tea 3gx10x8	3,773.	4,228.
10	A-61117	AM GSG Candy 1 lb / Bag	1,668.	1,878.
11	A-61150	AM GSG Candy 6 oz Bag	57,994.	61,244.
12	A-61220	AM GSG RT Tea 20 Bags	4,992.	4,999.
13	A-61243	AM GSG RT Tea 40 Bags	5,855.	6,415.
14	A-61260N	AM GSG RT Tea 60 Bags	4,245.	5,331.
15	A-61280	AM GSG RT Tea 80 Bags	0.	0.
16	F-20141	Ferrero Coll. 24 pcs (w/Hazelnut)#20148	800.	860.
17	F-12122	Ferrero Rocher 12 Pcs	612.	1,716.
18	F-12402	Ferrero Rocher 24 Pcs Diamond	1,502.	1,730.
19	F-12418	Ferrero Rocher 24 pcs Diamond Display(48)	8.	13.
20	F-12998	Ferrero Rocher 48's Flat Gift Box	0.	0.
21	F-50140	Gavottes Crepes 1250g #011120	281.	329.
22	F-50139	Gavottes Crepes 500g	3.	21.
23	F-76010	Kjeldsens 1 lb Danish Cookie(#98838)	8,388.	8,484.
24	F-00491	Loacker Wafer Hazelnut 250g(#10491)	14,314.	14,980.
25	F-97000	MX Eggrolls (32pcs)	1,152.	1,440.
26	F-00372	Loacker GP Bisc Noisette Choc 3.53oz(#10392)	1,164.	1,248.

A1 fx PO Number

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	PO Numbe	PO Date	Required D	Promised S	Receipt Dat	POP Receip	Item Numb	Item Descr	QTY Shippe	QTY Invoic	Unit Cost	Extended C	Vendor ID	Location C
2	37486	12/13/2022	12/29/2022	12/13/2022	1/11/2023	30635	D-04011	POP In Ging	940	940	59.358	55796.52	IM9HA8A	3
3	39795	8/2/2024	11/29/2024	11/3/2024	12/6/2024	35093	D-04011	POP In Ging	300	0	42.35	12705	IM9HA8A	1
4	40801	3/19/2025	7/24/2025	6/22/2025	7/31/2025	36807	D-04011	POP In Ging	420	0	42.35	17787	IM9HA8A	1
5	36827	4/27/2022	9/2/2022	4/27/2022	1/10/2023	30647	F-04112	POP Ginger	7836	7836	6.775	53089.9	IM9SI1A	2
6	36828	4/27/2022	9/5/2022	4/27/2022	2/13/2023	30738	F-04114	POP Ginger	7836	7836	6.421	50312.6	IM9SI1A	1
7	36834	4/27/2022	9/30/2022	4/27/2022	11/18/2022	30482	F-04111	POP Ginger	7500	7500	6.308	47306.7	IM9PT2A	2
8	36838	4/28/2022	10/21/2022	4/28/2022	12/14/2022	30576	F-04111	POP Ginger	7500	7500	5.285	39636.8	IM9PT2A	1
9	36840	4/28/2022	9/23/2022	4/28/2022	1/16/2023	30653	F-04112	POP Ginger	7836	7836	6.764	53004.9	IM9SI1A	2
10	36841	4/28/2022	10/23/2022	4/28/2022	11/2/2022	30424	F-04073	POP Ginger	4356	4356	9.661	42085.1	IM9SI1A	1
11	36842	4/28/2022	10/25/2022	4/28/2022	3/7/2023	30817	F-04112	POP Ginger	696	696	6.783	4720.97	IM9SI1A	2
12	36842	4/28/2022	10/25/2022	4/28/2022	3/7/2023	30817	F-04114	POP Ginger	4740	4740	6.783	32151.42	IM9SI1A	2
13	36933	5/27/2022	9/25/2022	5/27/2022	11/18/2022	30481	F-04075	POP Ginger	1550	1550	8.174	12669.7	IM9SI1A	1
14	36933	5/27/2022	9/25/2022	5/27/2022	11/18/2022	30481	F-04130	POP Ginger	1440	1440	16.098	23181.12	IM9SI1A	1
15	36954	6/6/2022	11/30/2022	6/6/2022	12/16/2022	30586	F-04111	POP Ginger	7500	7500	5.266	39491.8	IM9PT2A	1
16	36956	6/6/2022	12/30/2022	6/6/2022	1/12/2023	30646	F-04070	POP Ginger	4230	4230	8.611	36423.8	IM9PT2A	2
17	36960	6/6/2022	1/30/2023	6/6/2022	2/8/2023	30733	F-04070	POP Ginger	4230	4230	7.924	33516.5	IM9PT2A	1
18	36961	6/6/2022	2/5/2023	6/6/2022	2/3/2023	30719	F-04111	POP Ginger	7500	7500	5.212	39092.8	IM9PT2A	3
19	36962	6/6/2022	2/10/2023	6/6/2022	2/6/2023	30720	F-04111	POP Ginger	7500	7500	5.205	39039.3	IM9PT2A	1
20	36971	6/8/2022	11/25/2022	6/8/2022	12/13/2022	30577	F-04073	POP Ginger	4356	4356	8.089	35235.68	IM9SI1A	1
21	36972	6/8/2022	12/25/2022	6/8/2022	3/7/2023	30816	F-04114	POP Ginger	7836	7836	6.405	50192.6	IM9SI1A	1
22	36973	6/8/2022	12/30/2022	6/8/2022	1/9/2023	30627	F-04073	POP Ginger	4356	4356	8.608	37495.6	IM9SI1A	2
23	36974	6/8/2022	1/25/2023	6/8/2022	1/20/2023	30663	F-04073	POP Ginger	4356	4356	8.593	37431.11	IM9SI1A	2
24	37194	8/29/2022	1/10/2023	8/29/2022	1/20/2023	30666	F-04075	POP Ginger	2607	2607	7.965	20765.5	IM9SI1A	1
25	37194	8/29/2022	1/10/2023	8/29/2022	1/20/2023	30666	F-04130	POP Ginger	880	880	15.636	13759.68	IM9SI1A	1
26	37278	9/29/2022	12/5/2022	9/29/2022	12/12/2022	30560	F-04074	POP Ginger	4356	4356	8.032	34985.7	IM9SI1A	1
27	37403	11/11/2022	2/20/2023	2/20/2023	3/2/2023	30804	A-61117	AM GSG Ca	450	450	59.815	26916.75	IM9GO3A	1
28	37403	11/11/2022	2/20/2023	2/20/2023	3/2/2023	30804	A-61150	AM GSG Ca	60	60	39.014	2340.84	IM9GO3A	1
29	37419	11/17/2022	1/20/2023	11/17/2022	1/31/2023	30696	F-04074	POP Ginger	4356	4356	7.911	34458.2	IM9SI1A	1

Sales and available inventory: prediction of demand and **distribution**

Categorization through cosine similarity and embeddings with NLP



1. If cosine_similarity > 0.5, Flag opportunities for "develop a new PoP-branded product"

2. HuggingFace transformers library with fine-tuning on custom dataset